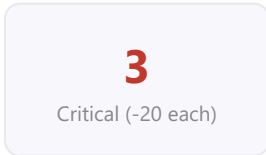
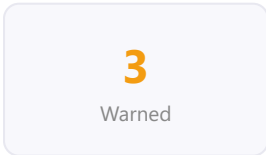
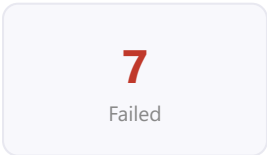


Data Contract Enforcer — Weekly Report

Generated: 20260402 | Pipeline: TenX Academy Week 7 | Contracts validated: 6 | Total checks: 96

1 • Data Health Score



Pipeline health is critically low at 30/100 (3 critical violation(s) active); the pipeline should be halted until root causes are resolved.

Formula: base = 89.6 (86/96 x 100) – 3 x 20 = 30

2 • Violations This Week

Count by Severity

Severity	Count
CRITICAL	3
HIGH	1
MEDIUM	3
WARNING	3

Top 3 Failing Checks

- [CRITICAL] LangSmith Traces - end_time > start_time: 150/150 violations. Affects 150 record(s). Impact: downstream time-series analytics and SLA calculations will produce incorrect results. Detail: 150 records violate 'end_time > start_time'.
- [CRITICAL] Week 1 – Intent Records - intent_id: 1 duplicates in 2 values. Affects 1 record(s). Impact: downstream consumers may surface incorrect data. Detail: 'intent_id' has 1 duplicate values. Duplicates indicate an idempotency violation in the pipeline.
- [CRITICAL] Week 5 – Events - recorded_at >= occurred_at: 100/100 violations. Affects 100 record(s). Impact: downstream time-series analytics and SLA calculations will produce incorrect results. Detail: 100 records violate 'recorded_at >= occurred_at'.

3 • Schema Changes Detected (Past 7 Days)

2 change(s) detected - 1 breaking, 1 compatible.

Dataset	Field	Change Type	Verdict	Plain Summary	Required Action
Week 3 – Extractions	review_notes	ADD_NULLABLE_COLUMN	COMPATIBLE	field 'review_notes' added (nullable) - no action needed	None. Downstream consumers can ignore the new column.
Week 3 – Extractions	doc_id	REMOVE_COLUMN	BREAKING	field 'doc_id' removed - any consumer reading this field will break	Deprecation period mandatory (minimum 2 sprints). Blast radius report required. Each affected consumer must acknowledge removal in writing (JIRA ticket or PR comment).

4 • AI System Risk Assessment

Overall AI Risk: LOW - Overall AI system risk: LOW. AI pipeline is operating within acceptable parameters.

- **Embedding Drift** [PASS]

Cosine distance from baseline centroid: 0.0000. Within acceptable range.
- **Prompt Input Schema** [PASS]

0/50 records quarantined (0.0%). Prompt inputs are well-formed.
- **Output Schema Violations** [PASS]

Violation rate=0.0% (threshold=2.0%). Trend vs baseline: stable. Rate within acceptable bounds.

5 • Recommended Actions

#1

CRITICAL

Review and address the breaking change for doc_id removal, ensuring all downstream consumers are notified and updated to handle the schema change.

Fix LangSmith trace ingestion to populate end_time correctly: update src/langsmith/ingest.py to set end_time = run.end_time (not null). All 150 traces currently have end_time ≤ start_time, breaking SLA dashboards.

#2

CRITICAL

Fix event timestamp population in the Week 5 event emitter / event emitter.py, fix

Fix event timestamp population in the Week 5 event emitter: update src/week5/event_emitter.py so recorded_at is set after occurred_at (currently 100/100 events violate this constraint), or swap the field assignment to use UTC clock at emit time.

#3

CRITICAL

Resolve duplicate intent_id in Week 1 intent records / intent_router.py, fix

Resolve duplicate intent_id in Week 1 intent records: update src/week1/intent_router.py to generate a new UUID per record instead of reusing 'INT-001'. Duplicate IDs break idempotency guarantees for downstream deduplication.